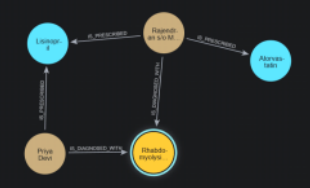
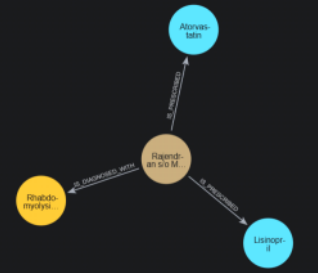


1	Find patients PRESCRIBED Lisinopril (or equivalent) who were DIAGNOSED WITH Rhabdomyolysis within 30 days of the prescription, and whose nodes or related clinical data mention 'myalgia' or 'muscle pain'.	Multi-hop path • Path: Medication (Lisinopril) → [GDS Similarity] → Medication (Related) → [IS, PRESCRIBED] → Patient → [IS, DIAGNOSED_WITH] → Condition (RDM) • Attributes Checked: [similarity score], [start_date], [occurrence_date], [clinical_summary: 'myalgia' OR 'muscle pain']	Expected Results 2 patients: Priya and Rajendran (clinical notes states 'soreness') Medications: Lisinopril and Atorvastatin (linked by similarity).	Actual results  <table data-bbox="767 358 1083 441"><thead><tr><th>Patient</th><th>Drug_Used</th><th>Score</th></tr></thead><tbody><tr><td>"Rajendran s/o Muthu"</td><td>"Atorvastatin"</td><td>0.75</td></tr><tr><td>"Rajendran s/o Muthu"</td><td>"Lisinopril"</td><td>1.0</td></tr><tr><td>"Priya Devi"</td><td>"Lisinopril"</td><td>1.0</td></tr></tbody></table>	Patient	Drug_Used	Score	"Rajendran s/o Muthu"	"Atorvastatin"	0.75	"Rajendran s/o Muthu"	"Lisinopril"	1.0	"Priya Devi"	"Lisinopril"	1.0	Remarks "equivalent" works using GDS similarity between drugs Priya direct signal Rajendran equivalent signal	MCP Query Only Rajendran								
Patient	Drug_Used	Score																								
"Rajendran s/o Muthu"	"Atorvastatin"	0.75																								
"Rajendran s/o Muthu"	"Lisinopril"	1.0																								
"Priya Devi"	"Lisinopril"	1.0																								
2	Find patients who were on a medication, experienced a Trigger (Statin-Induced Myopathy), and were switched to a different medication within 48 hours .	Multi-hop path • Path: Patient → [IS, PRESCRIBED] → Medication (Old) AND Patient → [IS, PRESCRIBED] → Medication (New) • Attributes Checked: [start_date (Old)], [start_date (New)], [occurrence_date (Trigger)], [time_diff: <= 48 hours]	Patient: Tan Wei Ling Was on Atorvastatin, experienced "Statin-Induced Myopathy", switched to Ezetimibe within 48hr	 <table data-bbox="767 710 1083 815"><thead><tr><th>patient</th><th>From_Drug</th><th>To_Drug</th><th>Relevance_Score</th></tr></thead><tbody><tr><td>"Tan Wei Ling"</td><td>"Atorvastatin"</td><td>"Ezetimibe"</td><td>0.5</td></tr><tr><td>"Tan Wei Ling"</td><td>"Atorvastatin"</td><td>"Ezetimibe"</td><td>0.5</td></tr><tr><td>"Tan Wei Ling"</td><td>"Lisinopril"</td><td>"Ezetimibe"</td><td>0.125</td></tr><tr><td>"Tan Wei Ling"</td><td>"Lisinopril"</td><td>"Ezetimibe"</td><td>0.125</td></tr></tbody></table>	patient	From_Drug	To_Drug	Relevance_Score	"Tan Wei Ling"	"Atorvastatin"	"Ezetimibe"	0.5	"Tan Wei Ling"	"Atorvastatin"	"Ezetimibe"	0.5	"Tan Wei Ling"	"Lisinopril"	"Ezetimibe"	0.125	"Tan Wei Ling"	"Lisinopril"	"Ezetimibe"	0.125	drug switch to a clinically equivalent one which is functionally or structurally similar	Works ✓ Patients who switched • Tan Wei Ling — id: P000000000 ◦ Prior medication: Atorvastatin ◦ Trigger: Statin-Induced Myopathy ◦ New medication: Ezetimibe • Tan Wei Ling — id: P000000000 ◦ Prior medication: Lisinopril ◦ Trigger: Statin-Induced Myopathy ◦ New medication: Ezetimibe
patient	From_Drug	To_Drug	Relevance_Score																							
"Tan Wei Ling"	"Atorvastatin"	"Ezetimibe"	0.5																							
"Tan Wei Ling"	"Atorvastatin"	"Ezetimibe"	0.5																							
"Tan Wei Ling"	"Lisinopril"	"Ezetimibe"	0.125																							
"Tan Wei Ling"	"Lisinopril"	"Ezetimibe"	0.125																							
3	Find cases where the PRESCRIBED duration for Atorvastatin and Lisinopril overlapped in time, followed by an occurrence_date for a Condition_RDM diagnosis.	Multi-hop path • Path: Patient → [IS, PRESCRIBED] → Medication (Atorvastatin) AND Patient → [IS, PRESCRIBED] → Medication (Lisinopril) → [Sequence] → Condition (RDM) • Attributes Checked: [start_date (Med 1)], [duration_days (Med 1)], [start_date (Med 2)], [duration_days (Med 2)], [occurrence_date (Diagnosis)]	Patient: Rajendran Drugs: Lisinopril and Atorvastatin, injury occurred exactly 14 days after the second drug was added	 <table data-bbox="767 1102 1195 1151"><thead><tr><th>Patient ID</th><th>ACE_Inhibitor</th><th>Statin</th><th>Injury_Date</th><th>ACE_Start</th><th>Statin_Start</th></tr></thead><tbody><tr><td>"Rajendran s/o M"</td><td>"Lisinopril"</td><td>"Atorvastatin"</td><td>2024-04-15</td><td>2024-01-15</td><td>2024-04-01</td></tr></tbody></table>	Patient ID	ACE_Inhibitor	Statin	Injury_Date	ACE_Start	Statin_Start	"Rajendran s/o M"	"Lisinopril"	"Atorvastatin"	2024-04-15	2024-01-15	2024-04-01	Shows drug overlap window, currently just native query filter and match	Works ✓ Overlap cases found • Patient: Rajendran s/o M ◦ Atorvastatin: start 2024-01-15 ◦ Lisinopril: start 2024-04-01 ◦ Overlap period: 2024-04-01 to 2024-04-15 ◦ Condition: Rhabdomyolysis								
Patient ID	ACE_Inhibitor	Statin	Injury_Date	ACE_Start	Statin_Start																					
"Rajendran s/o M"	"Lisinopril"	"Atorvastatin"	2024-04-15	2024-01-15	2024-04-01																					
4	Across the entire population, what is the incidence rate of RDM within 30 days of a patient being on a high-risk co-exposure (any ACE inhibitor + any Statin), and does a history of Hypertension significantly accelerate the time-to-injury? <i>Modified from "Sample Query" slide</i>	Multi-hop path • Path: Condition (Hypertension) → [HAS_HISTORY] → Patient → [IS, PRESCRIBED] → Medication (Lisinopril) → [Sequence] → Condition (RDM) • Attributes Checked: [occurrence_date (Hypertension)], [start_date (Med)], [occurrence_date (RDM)], [time_to_injury gap]	summary of the drug-drug interaction (DDI) risk Patient count + drug combo	 <table data-bbox="767 1424 1204 1503"><thead><tr><th>Drug_Combination</th><th>Patient_Count</th><th>Avg_Days_to_Injury</th><th>Patients_With_Hypertension</th></tr></thead><tbody><tr><td>"Lisinopril + Atorvastatin 1 or more"</td><td>14.0</td><td>1</td></tr><tr><td>"Atorvastatin + Lisinopril 1 or more"</td><td>14.0</td><td>1</td></tr></tbody></table>	Drug_Combination	Patient_Count	Avg_Days_to_Injury	Patients_With_Hypertension	"Lisinopril + Atorvastatin 1 or more"	14.0	1	"Atorvastatin + Lisinopril 1 or more"	14.0	1	GDS finds patients on drugs equivalent to the high risk pair Currently only Rajendran matches	Works ✓ Quick answer • Exposed patients (ACE inhibitor + Statin) • RDM cases within 30 days of overlap: 1 • Incidence: 1 / 2 = 50.0% (1: 500 per 1,000) △ Case summary (for the RDM cases) Patient: Rajendran s/o Muthu (P02) • All RDM cases (n=1) had prior Hypertension (distribution to compare)										
Drug_Combination	Patient_Count	Avg_Days_to_Injury	Patients_With_Hypertension																							
"Lisinopril + Atorvastatin 1 or more"	14.0	1																								
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5	Identifies patients who haven't been diagnosed with RDM yet, but whose clinical profile (conditions and history) is highly similar to patients who have been diagnosed RDM <i>Condition match</i>	Multi-hop path • Path: Patient (Diagnosed Case) → [IS, DIAGNOSED_WITH] → Condition → [IS, DIAGNOSED_WITH] → Patient (Not Yet Diagnosed) • Attributes Checked: [Jaccard similarity score], [clinical_summary (for profile matching)], [age], [gender]	Condition combo + patient match % + shared conditions	<table data-bbox="767 1509 1212 1632"><thead><tr><th>High_Risk_Patient</th><th>Clinical_Similarity</th><th>Match_Score</th><th>Shared_Conditions</th></tr></thead><tbody><tr><td>"Priya Devi"</td><td>"Rajendran s/o Muthu"</td><td>1.0</td><td>["hypertension", "Rhabdomyolysis"]</td></tr><tr><td>"Rajendran s/o Muthu"</td><td>"Priya Devi"</td><td>1.0</td><td>["hypertension", "Rhabdomyolysis"]</td></tr><tr><td>"Chloe de Souza"</td><td>"Rajendran s/o Muthu"</td><td>0.5</td><td>["hypertension"]</td></tr><tr><td>"David Lim"</td><td>"Rajendran s/o Muthu"</td><td>0.5</td><td>["hypertension"]</td></tr></tbody></table>	High_Risk_Patient	Clinical_Similarity	Match_Score	Shared_Conditions	"Priya Devi"	"Rajendran s/o Muthu"	1.0	["hypertension", "Rhabdomyolysis"]	"Rajendran s/o Muthu"	"Priya Devi"	1.0	["hypertension", "Rhabdomyolysis"]	"Chloe de Souza"	"Rajendran s/o Muthu"	0.5	["hypertension"]	"David Lim"	"Rajendran s/o Muthu"	0.5	["hypertension"]	Using Rajendran as base to compare	Works ✓ Quick finding Found multiple patients without RDM whose clinical profile closely matches that of Rajendran s/o Muthu (P02) Top conditions (patients ranked by Jaccard similarity) Candidate: 14 Chloe de Souza (P04) 000 David Lim (P08) 000 Farouk S. Ahmed (P07) 000 Muhammad Syamsul (P05) 000 Siti Aminah (P06) 000 Tan Wei Ling (P02) 000 (Others with lower scores 0.25-0.33) Method (brief) • Computed Jaccard similarity between patients • Compared non-RDM patients to RDM patients • Identifiable: reported and ranked by similarity (1: 100) Method (brief) Watchlist (top results) Patient: 14 Tan Wei Ling (P02) 51 Chloe de Souza (P04) 39 David Lim (P08) 40 Farouk S. Ahmed (P07) 40 Muhammad Syamsul (P05) 42 Siti Aminah (P06) 40 Note: many other patients have low scores / no action Suggested next steps • Check full records for CRITICAL/PROB patients • Review phenotype match with 10 CR + 100 • Export watchlist to CSV/JSON for review or share Would you like me to check full details for the CRITICAL/PROB patients?
High_Risk_Patient	Clinical_Similarity	Match_Score	Shared_Conditions																							
"Priya Devi"	"Rajendran s/o Muthu"	1.0	["hypertension", "Rhabdomyolysis"]																							
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"David Lim"	"Rajendran s/o Muthu"	0.5	["hypertension"]																							
6	Identify patients who share a clinical profile with known injury cases and are currently exposed to high-risk polypharmacy, but have not yet manifested RDM <i>Patients at risk of RDM</i>	Multi-hop path • Path: Patient (Candidate) → [GDS Similarity] → Patient (Case) AND Patient (Candidate) → [IS, PRESCRIBED] → Multiple Medications • Attributes Checked: [similarity score], [start_date (active prescriptions)], [duration_days], [clinical_summary (absence of RDM keywords)]	Patient + Risk summary % + flag Tan Wei Ling flagged as critical	<table data-bbox="767 1785 1184 1870"><thead><tr><th>Patient</th><th>Clinical_Similarity</th><th>Medication_Risk</th><th>Overall_Risk</th><th>Category</th></tr></thead><tbody><tr><td>"Tan Wei Ling"</td><td>0.5</td><td>0.5625</td><td>0.5375</td><td>"CRITICAL"</td></tr><tr><td>"Chloe de Souza"</td><td>0.5</td><td>0.125</td><td>0.275</td><td>"HIGH"</td></tr><tr><td>"David Lim"</td><td>0.5</td><td>0.125</td><td>0.275</td><td>"HIGH"</td></tr></tbody></table>	Patient	Clinical_Similarity	Medication_Risk	Overall_Risk	Category	"Tan Wei Ling"	0.5	0.5625	0.5375	"CRITICAL"	"Chloe de Souza"	0.5	0.125	0.275	"HIGH"	"David Lim"	0.5	0.125	0.275	"HIGH"	Combines and weights Case 4&5	Works abt, need further prompting Watchlist (top results) Patient: 14 Tan Wei Ling (P02) 51 Chloe de Souza (P04) 39 David Lim (P08) 40 Farouk S. Ahmed (P07) 40 Muhammad Syamsul (P05) 42 Siti Aminah (P06) 40 Note: many other patients have low scores / no action Suggested next steps • Check full records for CRITICAL/PROB patients • Review phenotype match with 10 CR + 100 • Export watchlist to CSV/JSON for review or share Would you like me to check full details for the CRITICAL/PROB patients?
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"David Lim"	0.5	0.125	0.275	"HIGH"																						

, age: 51
 atorvastatin (start: 2024-02-01, duration_days: 5)
 Myocardial Infarction (occurred: 2024-02-06)
 simvastatin (new start: 2024-02-07) ☒
 , age: 51
 lisinopril (start: 2024-01-15, duration_days: 180)
 Myocardial Infarction (occurred: 2024-02-06)
 simvastatin (new start: 2024-02-07) ☒

Similarity	Medication Risk	Overall Risk	Category	Action/Recent Mode
0.50	0.50	0.50	CRITICAL	(Accurate, Latency, Error)
0.50	0.505	0.275	HIGH	JoinOp[0]
0.50	0.505	0.275	HIGH	JoinOp[0]
0.50	0.505	0.275	HIGH	JoinOp[0]
0.50	0.505	0.275	HIGH	JoinOp[0]
0.50	0.505	0.275	HIGH	JoinOp[0]

Quick Notes Page 2